BRAC UNIVERSITY

Inspiring Excellence

CSE340

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Assignment 4

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Section : 8

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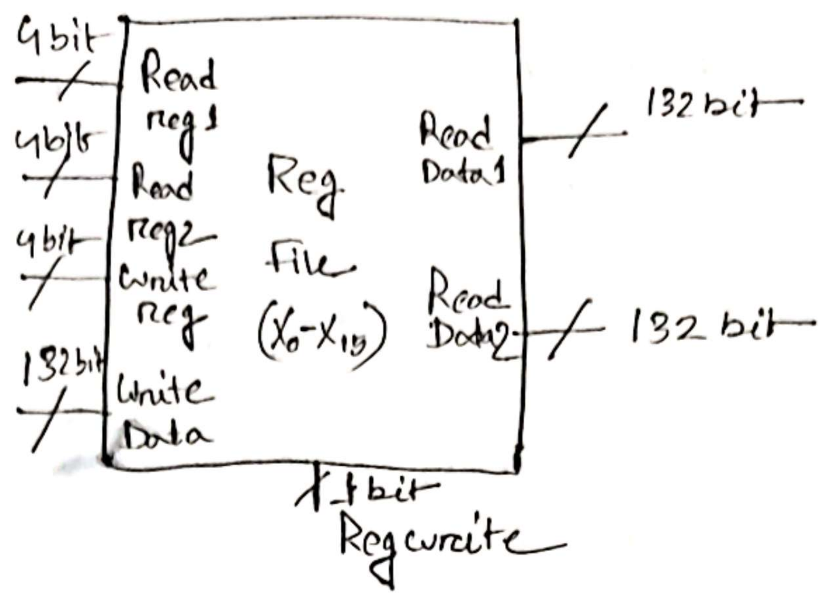
1.



2. ALU control unit takes Ins [30, 14-12] bit cause ~~this~~ these bits indicates func3 and func7 [5] bit.

From the opcode 7 bits ALU understand weather the operation is R-type, I-type, Load, store, branch etc. But, with func3 bits it understand the basic operation like, add, sub, XOR, OR, And etc. for more distinguished between simillar operation like and sub it uses func7 [5] bit.

3) Zero pin of the ALU indicates whether the result is zero or not. It is crucial for branch decisions like BEQ/BNE efficiently. Cause it directly signals between two registers.

4) For single cycle

Clock cycle ⇒ $(20+20+40+20) = 100s$

For pipeline,

$(40s \times 5) \Rightarrow 200s$

∴ Single-cycle ~~sto~~ setup will run the instruction faster.

7) A control Hazard happend in pipeline when the CPU doesn't ~~Kou~~ know yet which instruction should run next. Because it is still figuring out the outcome of the branch on jump instruction. Because, In a pipeline instruction are fetched before the result of the current one for a branch the cpu needs to check the conditions. That decision is usually resolved in the Execution stage byt by then, the Instruction fetch stage may have already fetched the wrong instruction.

8) To avoid structural hazard, In a pipeline one stage is fetching and instruction while another stage may need to access data at the same time. If instruction and data share same memory both component need to access at a single stage which

is a structural hazard. That's why instruction memory and data memory kept seperate.

6) LD x_5, 10(x_{21})

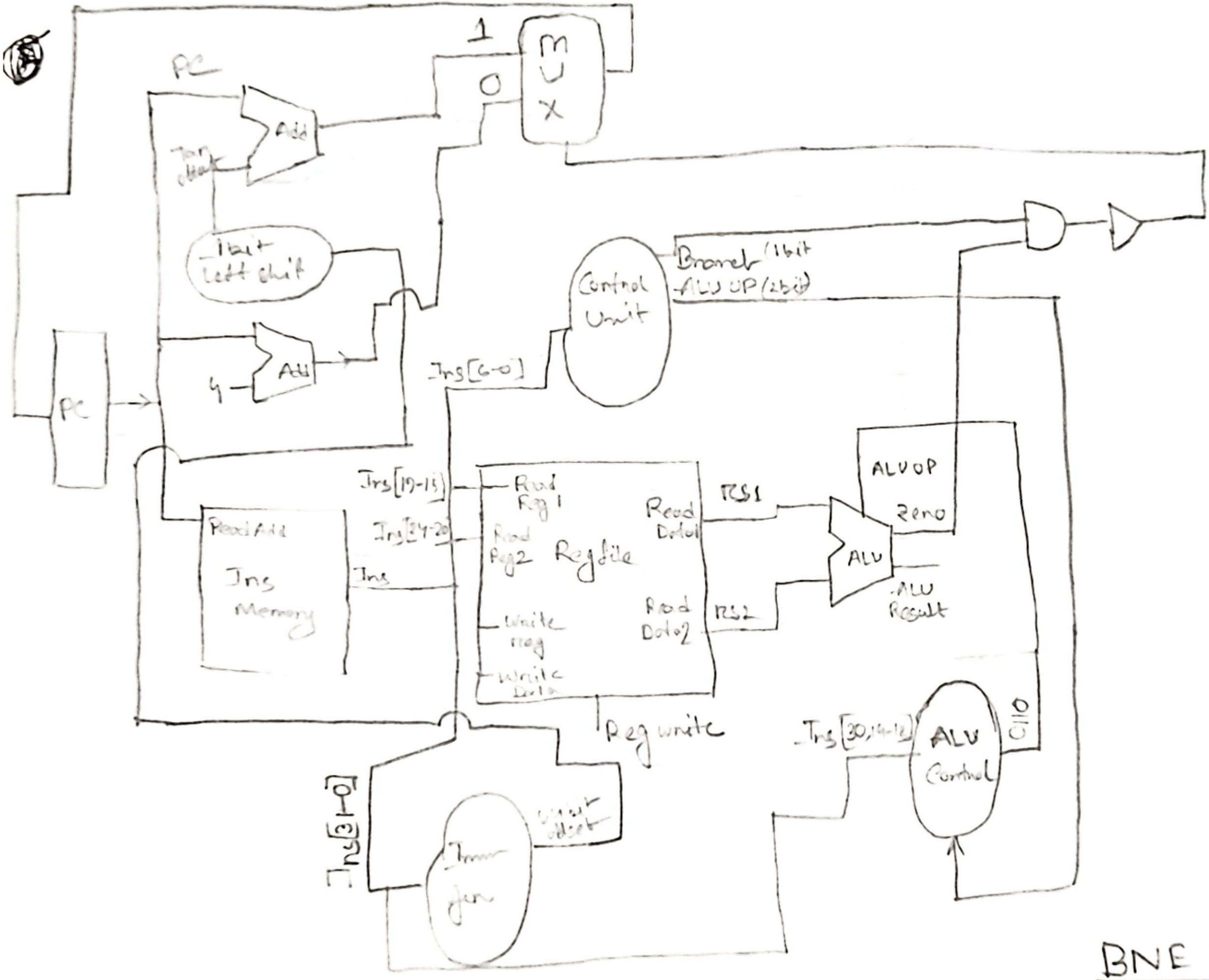
ALU Source = 0

~~Right~~ Regwrite = 1

MemRead = 1

Mem Write = 0

MemtoReg = 0



BNE Datapath

In here, a Not gate implemented after ~~the~~ the AND Grate

so when $rs1 \neq rs2$; ~~ALO~~ ALU zero pin will give the signal 0 and ~~Branch~~ for branch signal it will give 1 for branching.

$0 \cdot 1 \Rightarrow 0$ and not gate will inverse the value into 1 ; which is a selector signal for MUX and in PIN no 1 is PC+offset in MUX so, It will jump into the label even after $rs1 \neq rs2$ which mean it is working for BNE instruction.